Bi-level Adversarial Reinforcement Learning for Robust Network Threat Detection

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***Abstract*—Network threat detection faces significant challenges from adaptive attackers, dataset shift, and evasion-oriented perturbations. To address these issues, this paper studies a Bi-level Adversarial Reinforcement Learning (Bi-ARL) framework in which a defender policy is optimized against an adaptive attacker through a nested leader-follower game. We implement the attacker-defender interaction as a bi-level training process and evaluate it on three public tabular intrusion-detection benchmarks: NSL-KDD, UNSW-NB15, and CIC-IDS2017. On NSL-KDD, Bi-ARL achieves the best mean F1 score among the reinforcement learning variants (80.17%) and substantially reduces false positives relative to vanilla PPO (10.41% vs. 39.57%). However, the modern-dataset results are notably weaker. On UNSW-NB15, supervised baselines remain clearly superior, with HGBT-IDS reaching 89.45% mean F1 and 26.31% FPR, while the RL methods exhibit high variance across seeds. On CIC-IDS2017 under a random stratified split, strong tabular baselines nearly saturate the benchmark with about 99.7% mean F1, whereas short-run RL baselines remain far behind. These findings indicate that the present Bi-ARL design improves the clean-data precision-recall trade-off in a controlled benchmark, but does not yet provide stable modern cross-dataset robustness.**

***Index Terms*—Network Threat Detection, Adversarial Reinforcement Learning, Bi-level Optimization, Intrusion Detection Systems, Game Theory.**

I. Introduction

Network security remains a central challenge for modern computing systems. Traditional intrusion detection systems (IDS) rely heavily on signatures or static statistical rules, which limits their ability to cope with zero-day attacks, distribution shift, and adversarial manipulation [1]–[3]. Recent learning-based IDS research has therefore moved toward self-supervised generalization [4], [5], open-set recognition [6], and transformer-based feature modeling [7]. At the same time, recent evidence shows that strong in-distribution benchmark results can degrade sharply under cross-dataset testing or stronger perturbations [3], [8]. Reinforcement learning (RL) offers a natural route toward adaptive defense, but many existing RL-based IDS pipelines still train against either static disturbances or simultaneously updated opponents, without explicitly approximating an attacker's best response.

This paper proposes a **Bi-level Adversarial Reinforcement Learning (Bi-ARL)** framework to address this gap. Unlike standard multi-agent RL, where both agents are optimized simultaneously, our approach models the interaction as a **Stackelberg game**: the defender acts as the leader and is updated against an attacker that is repeatedly adapted toward a stronger response. The purpose is not to claim a solved robust IDS, but to test whether explicit leader-follower optimization yields a better detection trade-off than ordinary RL baselines.

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A. Contributions

Our main contributions are:

1) We formulate IDS training as a bi-level attacker-defender optimization problem and implement a practical PPO-based Bi-ARL pipeline for tabular network-flow benchmarks.
2) We show on NSL-KDD that Bi-ARL achieves the best mean F1 score among the RL baselines while substantially reducing FPR relative to vanilla PPO.
3) We extend the evaluation to UNSW-NB15 and CIC-IDS2017, where the results reveal a clear gap between the current RL formulation and strong supervised tabular baselines such as HGBT-IDS, XGBoost-IDS, LightGBM-IDS, and LSTM-IDS.
4) We provide a deliberately conservative empirical analysis: the proposed method is useful as an RL research prototype, but its robustness and cross-dataset stability remain limited, which identifies concrete priorities for the next iteration of the work.

II. Related Work

A. Intrusion Detection Systems

Traditional IDSs rely on signature matching, which remains effective for known attack templates but fails on zero-day or rapidly evolving threats. Recent surveys show that learning-based IDSs now span classical machine learning, deep neural networks, graph-based methods, and large-scale IoT-oriented pipelines [1], [2]. At the same time, dataset quality remains a major bottleneck: recent reviews emphasize that many public NIDS benchmarks contain outdated traffic patterns, shortcut features, or split assumptions that can overestimate deployment performance [9]. This issue is directly relevant to our work because the empirical conclusions of Bi-ARL change noticeably across NSL-KDD, UNSW-NB15, and CIC-IDS2017.

B. Generalization and Modern Deep IDS Models

An important recent trend is to move beyond single-dataset closed-set evaluation. Golchin *et al.* proposed SSCL-IDS, a self-supervised contrastive approach designed to improve detection under unseen traffic distributions [4]. Xu *et al.* further extended self-supervised learning to graph neural network based flow modeling, showing that representation learning can reduce the dependence on expensive labels while

remaining competitive with supervised approaches [5]. On the architecture side, transformer-based IDS models have also shown strong in-distribution performance; for example, IDS-MTran uses multi-scale branches and transformer backbones to obtain high accuracy on NSL-KDD, UNSW-NB15, and CIC-DDoS2019 [7]. These works define a stronger modern comparison set than legacy RL-only baselines, but they still generally optimize static prediction models rather than explicitly incorporating an adaptive attacker into training.

C. Robustness, Open-Set Detection, and Cross-Dataset Evaluation

Robustness has emerged as a separate research axis from plain accuracy. Wang *et al.* argue that adversarial perturbations and distribution shift should be treated as first-class evaluation criteria for ML-based NIDSs, rather than as optional stress tests [3]. Cantone *et al.* provide direct evidence that models with strong within-dataset performance can deteriorate severely under cross-dataset testing, highlighting the gap between benchmark success and deployment reliability [8]. In parallel, VAEMax tackles open-set intrusion detection by combining OpenMax with a variational autoencoder to identify previously unseen attacks on CIC-IDS2017 and CSE-CIC-IDS2018 [6]. These results motivate our focus on adaptive attacker-defender interaction, but they also imply that robustness claims should be made conservatively and backed by explicit protocol-sensitive evaluation.

D. Adversarial Reinforcement Learning

Reinforcement learning applied to security games allows for adaptive defense strategies and naturally accommodates sequential attacker-defender interaction. However, many practical RL-based IDS formulations still update the defender against either a static disturbance model or a simultaneously trained opponent, which can blur the distinction between ordinary multi-agent co-adaptation and a genuine best-response adversary. Our work focuses on this gap: rather than only using adversarial interaction as data augmentation, we formulate training as a nested optimization process in which the defender is optimized against an attacker that is repeatedly adapted toward a stronger response. This places the method closer to a Stackelberg-style robust control perspective than to standard simultaneous-update MARL. Empirically, however, our results also show that this formulation alone does not guarantee competitive performance on modern datasets, which is an important difference from stronger supervised tabular baselines.

III. Methodology

This section details the proposed Bi-level Adversarial Reinforcement Learning (Bi-ARL) framework. We first formalize the network threat detection problem as a Stackelberg game, then describe the bi-level optimization algorithm with PPO-based policy updates.

A. Problem Formulation

Consider a network environment with state space $\mathcal{S}$ representing tabular traffic-flow features. In our implementation, the state dimensionality depends on the dataset profile (e.g., 41 for NSL-KDD, 42 for UNSW-NB15, and 78 for CIC-IDS2017 after preprocessing). The interaction involves two agents: an Attacker ($\mathcal{A}$) and a Defender ($\mathcal{D}$).

Let $s_t \in \mathcal{S}$ denote the state at time t. The Attacker selects action $a_t^{\mathcal{A}} \in \mathcal{A}^{\mathcal{A}}$ (perturbation strategies), and the Defender selects $a_t^{\mathcal{D}} \in \mathcal{A}^{\mathcal{D}}$ (classification decision).

We model the problem as a **Stackelberg game**:

$$\max_{\theta_{\mathcal{D}}} J(\theta_{\mathcal{D}}, \theta_{\mathcal{A}}^*) = \mathbb{E}_{\tau \sim \pi_{\mathcal{D}}, \pi_{\mathcal{A}}^*} \left[\sum_{t=0}^{T} \gamma^t R_{\mathcal{D}}(s_t, a_t^{\mathcal{D}}, a_t^{\mathcal{A}*}) \right] \quad (1)$$

$$\text{s.t.} \quad \theta_{\mathcal{A}}^* = \arg\max_{\theta_{\mathcal{A}}} \mathbb{E}_{\tau \sim \pi_{\mathcal{A}}} \left[\sum_{t=0}^{T} \gamma^t R_{\mathcal{A}}(s_t, a_t^{\mathcal{D}}, a_t^{\mathcal{A}}) \right] \quad (2)$$

where $\gamma = 0.99$ is the discount factor, and τ denotes a trajectory.

B. Reward Functions

1) Attacker Reward: The Attacker aims to maximize evasion success:

$$R_{\mathcal{A}} = \begin{cases} +2.0 & \text{if FN (attack evades detection)} \\ -1.0 & \text{if TP (attack detected)} \end{cases} \quad (3)$$

2) Defender Reward: The Defender prioritizes detecting attacks while minimizing false alarms:

$$R_{\mathcal{D}} = \begin{cases} +1.0 & \text{if TP (correct detection)} \\ +0.5 & \text{if TN (correct rejection)} \\ -1.0 & \text{if FP (false alarm)} \\ -2.0 & \text{if FN (missed attack)} \end{cases} \quad (4)$$

In the codebase, these rewards are implemented through dataset-specific profiles. The NSL-KDD setting uses the values above directly, while UNSW-NB15 and CIC-IDS2017 use slightly stronger false-positive penalties to reduce immediate policy collapse under higher class imbalance and harder generalization settings. The common design principle is unchanged: false negatives are penalized more heavily than true negatives are rewarded, and false positives incur a meaningful deployment cost.

C. PPO-Based Policy Optimization

Both agents use Proximal Policy Optimization (PPO) [10]. Given policy $\pi_\theta(a|s)$, PPO maximizes the clipped surrogate objective:

$$L^{CLIP}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right] \quad (5)$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is the probability ratio, $\hat{A}_t$ is the advantage estimate, and $\epsilon = 0.2$ is the clipping parameter. The advantage is computed via Generalized Advantage Estimation (GAE) [11]:

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}, \quad \delta_t = r_t + \gamma V(s_{t+1}) - V(s_t) \tag{6}$$

with $\lambda = 0.95$ for bias-variance trade-off.

D. Bi-level Optimization Algorithm

1) Inner Loop: Attacker Convergence: The Attacker optimizes against a **fixed** Defender policy $\pi_{\mathcal{D}}^{\text{fixed}}$ until convergence. Convergence is measured by KL divergence:

$$D_{KL}(\pi_{\mathcal{A}}^{\text{old}} \| \pi_{\mathcal{A}}^{\text{new}}) = \mathbb{E}_{s,a}\left[\log \frac{\pi_{\mathcal{A}}^{\text{old}}(a|s)}{\pi_{\mathcal{A}}^{\text{new}}(a|s)}\right] < \tau_{KL} \tag{7}$$

where $\tau_{KL} = 0.01$. The Inner Loop terminates when $D_{KL} < \tau_{KL}$ or after K_{inner} steps. In the reproduced experiments, $K_{\text{inner}} = 5$ for NSL-KDD and $K_{\text{inner}} = 1$ for the harder modern datasets, reflecting a practical stabilization choice rather than a theoretically ideal setting.

2) Outer Loop: Defender Robustification: Once the Attacker converges to $\theta_{\mathcal{A}}^*$, the Defender updates against this worst-case adversary. The Attacker remains in `eval()` mode with frozen gradients during this phase.

Algorithm 1 Bi-level Adversarial RL (Bi-ARL)

Input: Episodes N, inner steps K_{inner}, KL threshold τ_{KL}
Initialize $\theta_{\mathcal{D}}, \theta_{\mathcal{A}}$ randomly
for episode $e = 1$ to N **do**
  // Inner Loop: Attacker Best-Response
  $\theta_{\mathcal{A}}^{\text{old}} \leftarrow \theta_{\mathcal{A}}$
  for $k = 1$ to K_{inner} **do**
    Collect trajectories $\tau \sim \pi_{\mathcal{A}}(\cdot|\theta_{\mathcal{A}}), \pi_{\mathcal{D}}^{\text{fixed}}$
    Update $\theta_{\mathcal{A}} \leftarrow \theta_{\mathcal{A}} + \alpha \nabla_{\theta_{\mathcal{A}}} L^{CLIP}(\theta_{\mathcal{A}})$
    Compute $D_{KL}(\pi_{\mathcal{A}}^{\text{old}} \| \pi_{\mathcal{A}})$
    if $D_{KL} < \tau_{KL}$ **then**
      break
      ▷ Attacker converged
    end if
  end for
  $\theta_{\mathcal{A}}^* \leftarrow \theta_{\mathcal{A}}$
  ▷ Freeze optimal Attacker

  // Outer Loop: Defender Robustification
  Collect trajectories $\tau \sim \pi_{\mathcal{A}}^*(\cdot|\theta_{\mathcal{A}}^*), \pi_{\mathcal{D}}(\cdot|\theta_{\mathcal{D}})$
  Update $\theta_{\mathcal{D}} \leftarrow \theta_{\mathcal{D}} + \beta \nabla_{\theta_{\mathcal{D}}} L^{CLIP}(\theta_{\mathcal{D}})$
end for
Return: Defender policy $\pi_{\mathcal{D}}(\cdot|\theta_{\mathcal{D}})$

E. Theoretical Justification

Our bi-level structure approximates a Stackelberg equilibrium, where the Defender's strategy is optimized against a stronger attacker response than in standard simultaneous-update MARL. In principle, this provides a more robust training objective than ordinary co-adaptation. In practice, however, the empirical sections show that convergence of the inner problem remains imperfect on harder datasets, so the current work should be interpreted as an operational approximation to Stackelberg-style robust training rather than a proof of worst-case security guarantees.

IV. Experiments

A. Experimental Setup

1) Datasets: We evaluate on two public tabular IDS benchmarks.

- **NSL-KDD**: `KDDTrain+` (125,973 records) and `KDDTest+` (22,544 records), with 41 input features.
- **UNSW-NB15**: official training split (175,341 records) and testing split (82,332 records), with 42 input features after removing identifier and auxiliary columns.

NSL-KDD remains useful for controlled ablation, but it is not sufficient as a standalone modern benchmark. We therefore include UNSW-NB15 to partially address the dataset obsolescence and cross-dataset concerns discussed in recent IDS literature [8], [9].

2) Baselines: We compare Bi-ARL with four baselines:

1) **Vanilla PPO**: single-agent RL without adversarial training.
2) **MARL**: simultaneous attacker-defender updates without bi-level nesting.
3) **LSTM-IDS**: a two-layer LSTM supervised baseline.
4) **HGBT-IDS**: histogram gradient boosting on tabular features.

These baselines still do not cover the strongest recent SSL, graph, or transformer IDS families [4], [5], [7], but they provide a materially stronger comparison set than the original half-finished repository.

3) Training Details: Unless otherwise noted, we report means over four random seeds (42, 123, 3407, 8888). NSL-KDD experiments use 100 RL episodes. For UNSW-NB15, we reduce the attacker perturbation magnitude and use a shorter bi-level inner loop in order to avoid immediate policy collapse during training; specifically, Bi-ARL uses one inner attacker update per outer step on UNSW-NB15, whereas the NSL-KDD setting uses five. LSTM-IDS is trained for 20 epochs, and HGBT-IDS uses a depth-8 histogram gradient boosting classifier with 300 boosting iterations.

B. Evaluation Metrics

- **Recall**: $TP/(TP + FN)$.
- **Precision**: $TP/(TP + FP)$.
- **F1-score**: harmonic mean of Precision and Recall.
- **False Positive Rate (FPR)**: $FP/(FP + TN)$.

C. Main Results

Table I reports the NSL-KDD results. Bi-ARL achieves the best mean F1 among the RL methods and clearly improves over vanilla PPO on FPR.

The NSL-KDD evidence supports a narrow claim: Bi-ARL offers a better precision-recall balance than vanilla PPO and slightly outperforms MARL on mean F1, but it does not dominate the supervised baseline on FPR.

TABLE I
NSL-KDD MAIN RESULTS (MEAN ACROSS 4 SEEDS)

Method	Recall	Precision	F1	FPR
Vanilla PPO	**76.39%**	79.75%	74.12%	39.57%
MARL	70.97%	91.65%	79.32%	9.54%
LSTM-IDS	65.05%	**96.73%**	77.78%	**2.91%**
Bi-ARL (Ours)	72.38%	90.56%	**80.17%**	10.41%

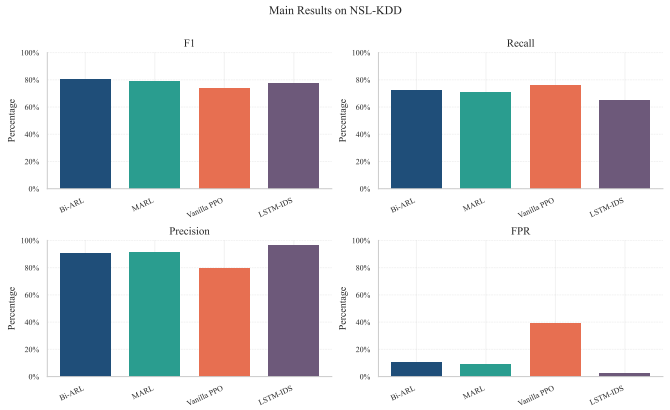

Fig. 1. Main-result visualization on NSL-KDD. Error bars indicate the standard deviation across four seeds.

Table II presents the more challenging UNSW-NB15 results. Here, the ranking changes substantially. HGBT-IDS and LSTM-IDS are both clearly stronger and more stable than the RL variants. Even after dataset-specific stabilization, Bi-ARL remains highly sensitive to initialization.

TABLE II
UNSW-NB15 MAIN RESULTS (MEAN ACROSS 4 SEEDS)

Method	Recall	Precision	F1	FPR
Vanilla PPO	66.21%	58.67%	59.42%	58.72%
MARL	74.98%	68.30%	57.26%	66.81%
LSTM-IDS	97.85%	78.82%	87.30%	32.26%
HGBT-IDS	98.30%	**82.07%**	**89.45%**	**26.31%**
Bi-ARL (Ours)	66.56%	48.80%	54.13%	53.56%

Cross-dataset interpretation: the current Bi-ARL implementation should not be presented as a state-of-the-art IDS. It is competitive only within the RL subset on NSL-KDD, while the UNSW-NB15 experiments show that stronger supervised tabular baselines remain substantially better in both effectiveness and stability.

D. Additional Benchmark: CIC-IDS2017

We further evaluate on CIC-IDS2017 in two settings. The first is a random stratified split (‘cic-ids2017-random‘), which is suitable for a conventional supervised-learning main table. The second is a stricter day-based split (‘cic-ids2017‘), where Monday–Thursday are used for training and Friday is held out for testing; this version is better interpreted as a cross-day generalization stress test.

Table III reports the random-split results. Here, all three tree-based baselines nearly saturate the benchmark, while LSTM-IDS remains slightly weaker but still very strong. Short-run RL smoke tests on the same split are far below

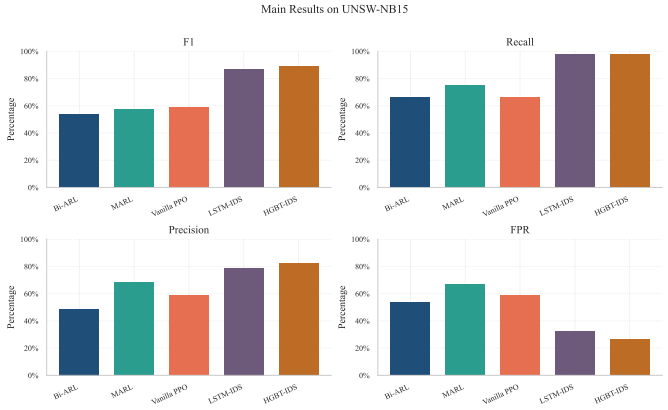

Fig. 2. Main-result visualization on UNSW-NB15. The supervised baselines remain clearly ahead of the current RL variants.

this level (Bi-ARL seed42: 36.48% F1; Vanilla PPO seed42: 33.17% F1), so we do not treat them as full main-table competitors for this dataset at the current stage.

TABLE III
CIC-IDS2017 RANDOM STRATIFIED SPLIT

Method	Recall	Precision	F1	FPR
LSTM-IDS	97.76%	92.27%	94.93%	2.04%
HGBT-IDS	99.85%	99.63%	99.74%	**0.09%**
XGBoost-IDS	**99.89%**	99.59%	99.74%	0.10%
LightGBM-IDS	99.87%	**99.62%**	**99.75%**	**0.09%**

The random-split results are useful because they show that the feature representation itself is highly learnable by modern tabular methods. At the same time, the near-saturated scores also imply that this split is not sufficient by itself to demonstrate robustness or distribution-shift resistance.

On the stricter day-based split, the picture changes sharply. XGBoost-IDS achieves only 38.26% mean F1, albeit with an extremely low FPR of 0.02%, while HGBT-IDS and LightGBM-IDS are even weaker. This confirms that CIC-IDS2017 is highly sensitive to the evaluation protocol. For this reason, the most defensible paper structure is to use the random split as a standard benchmark table and the day-based split as a harder generalization case study rather than mixing the two into a single headline claim.

E. Ablation Study

For NSL-KDD, the ablation results indicate that the main practical gain comes from adversarially informed training: removing the attacker raises clean-data FPR from 10.41% to 39.57%. The full Bi-ARL model also achieves the best mean F1 among the RL variants.

TABLE IV
NSL-KDD ABLATION RESULTS (MEAN ACROSS 4 SEEDS)

Variant	Recall	F1	FPR
w/o Inner Loop	70.98%	78.78%	10.82%
w/o Attacker (Vanilla PPO)	**76.39%**	74.12%	39.57%
w/o Bi-level (MARL)	70.97%	79.32%	**9.54%**
Full Bi-ARL	72.38%	**80.17%**	10.41%

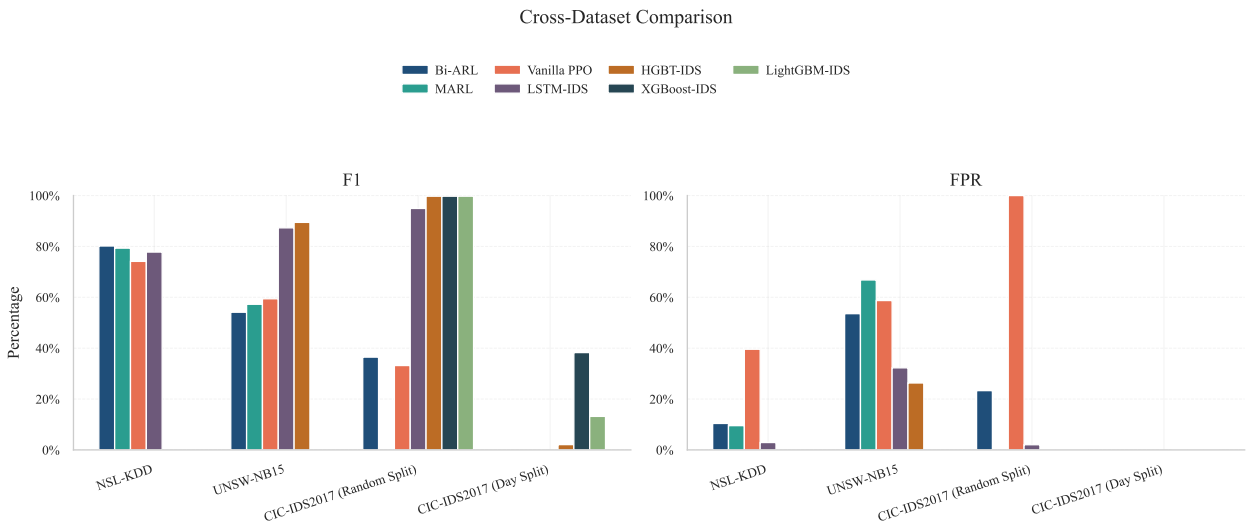

Fig. 3. Cross-dataset comparison of mean F1 and FPR. The figure summarizes the core message of the paper: Bi-ARL is competitive only on the controlled NSL-KDD benchmark, while modern tabular baselines dominate on UNSW-NB15 and CIC-IDS2017.

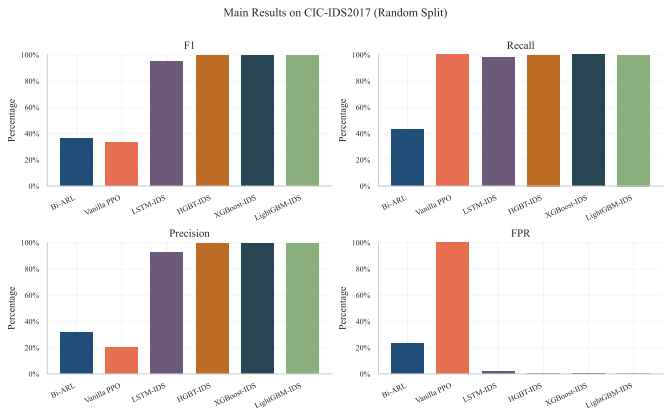

Fig. 4. Main-result visualization on CIC-IDS2017 under a random stratified split. Modern tree-based tabular baselines nearly saturate this protocol.

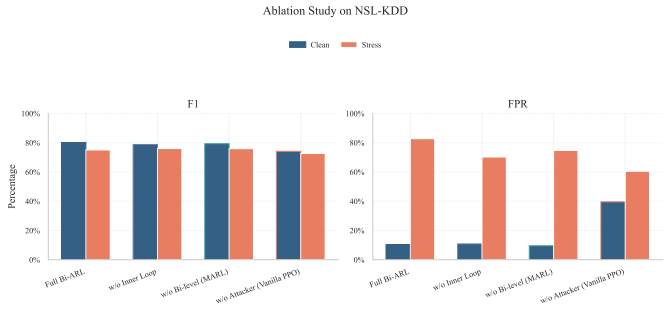

Fig. 5. Ablation comparison on NSL-KDD. The full model has the best clean-data F1 among the RL variants, while removing adversarial training sharply increases FPR.

On UNSW-NB15, however, the ablation trend is much less favorable. The fixed-attacker variant achieves lower mean FPR than full Bi-ARL (37.45% vs. 53.56%), and MARL attains higher recall at the cost of extremely high false alarms. This pattern indicates that the current inner-loop design is not yet

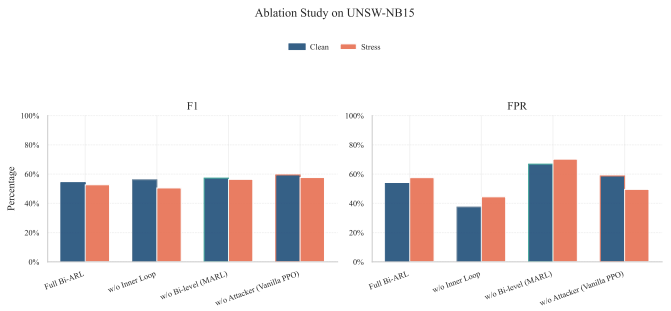

Fig. 6. Ablation comparison on UNSW-NB15. The favorable NSL-KDD trend does not transfer cleanly to the modern benchmark.

stable enough to justify a strong causal claim on modern data.

TABLE V
UNSW-NB15 ABLATION RESULTS (MEAN ACROSS 4 SEEDS)

Variant	Recall	F1	FPR
w/o Inner Loop	62.95%	56.07%	**37.45%**
w/o Attacker (Vanilla PPO)	66.21%	**59.42%**	58.72%
w/o Bi-level (MARL)	**74.98%**	57.26%	66.81%
Full Bi-ARL	66.56%	54.13%	53.56%

F. Adversarial Robustness on NSL-KDD

We evaluate FGSM attacks [12] on NSL-KDD with perturbation budget $\epsilon \in \{0.1, 0.3\}$:

$$s' = s + \epsilon \cdot \text{sign}(\nabla_s L(\theta_{\mathcal{D}}, s, y)). \tag{8}$$

Key finding: the present Bi-ARL model is only partially robust. Under mild attacks, it still preserves lower FPR than vanilla PPO, but stronger perturbations cause severe degradation, including near-universal false alarms at $\epsilon = 0.3$.

TABLE VI
NSL-KDD ADVERSARIAL ROBUSTNESS UNDER FGSM

Model	Attack	Recall	FPR
Bi-ARL	Clean	72.38%	10.41%
	FGSM-ϵ0.1	89.53%	47.03%
	FGSM-ϵ0.3	94.85%	100.00%
Vanilla PPO	Clean	76.39%	39.57%
	FGSM-ϵ0.1	81.64%	54.06%
	FGSM-ϵ0.3	86.94%	76.62%

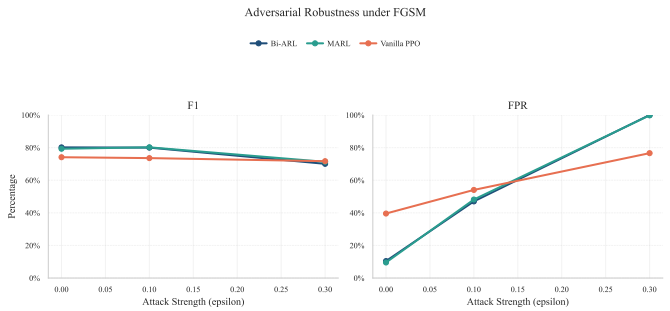

Fig. 7. FGSM robustness curves on NSL-KDD. Both RL models degrade as the perturbation budget increases, and Bi-ARL does not remain robust under stronger attacks.

G. Discussion

Taken together, the experiments support a cautious conclusion. Bi-ARL is a meaningful RL formulation for adversarial IDS training and improves over a naive PPO baseline on NSL-KDD. However, the modern-data results reveal substantial instability and a clear gap to strong supervised tabular methods. For submission-quality follow-up work, the most important next step is to stabilize the inner-loop optimization and add at least one more contemporary benchmark such as CIC-IDS2017 or CSE-CIC-IDS2018.

V. CONCLUSION

This paper introduced a Bi-level Adversarial Reinforcement Learning (Bi-ARL) framework for network threat detection. By modeling the attacker-defender interaction as a nested leader-follower game, the method goes beyond static supervised learning and standard simultaneous-update RL. In our reproduced NSL-KDD experiments, Bi-ARL achieves the best mean F1 score among the RL baselines and substantially lowers FPR compared with vanilla PPO, which confirms that adversarially informed policy optimization can improve the clean-data detection trade-off in a controlled setting. However, the broader evaluation across UNSW-NB15 and CIC-IDS2017 leads to a more restrained conclusion: strong supervised tabular baselines remain clearly stronger and more stable than the current RL formulation, and the relative ranking is highly sensitive to the evaluation protocol. The current system should therefore be viewed as a research prototype that demonstrates a useful RL perspective, rather than as a submission-ready robust IDS. The most important next steps are to stabilize the inner-loop optimization, reduce seed sensitivity, strengthen modern baseline coverage, and extend the evaluation to harder real-world protocols and additional public datasets such as CSE-CIC-IDS2018.

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